

A dictionary of quantum shifts in first-order gauged matrix models:

ADHM, Grassmannians and the Jordan quiver

AI* Masaya Kameyama

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Abstract

First-order gauged matrix quantum mechanics on S^1 computes equivariant indices of line bundles on the Higgs branch, but the identification of the Chern–Simons level depends on how the theory is regularised. Different papers report different “quantum shifts”: $k' \rightarrow k' - N$ for the ADHM model of the 4d quantum Hall effect [1], $k + n \mapsto k$ for the Chern–Simons matrix model on the framed Jordan quiver [2], and $k \rightarrow k + 1$ for the Polychronakos matrix model [3]. We give a single rule that accounts for all of them. Write the one-loop factor of every field mode of weight x and gauge fugacity w as $x^{s_\varepsilon} w^{s_w} / (1 - xw)$. Then the partition function is $Z = (\prod x^{s_\varepsilon}) \chi(M, \mathcal{O}(k' - \sum \text{charge} \cdot s_w))$, where M is the Higgs branch. It equals the twisted index $\chi(M, \mathcal{O}(k') \otimes K^s)$ if and only if $s_\varepsilon = s_w = s$ for all fields. Zeta-function regularisation ($s = \frac{1}{2}$) gives the Dirac (spin^c) index; gauge-covariant time lattices have $s_w = 0$ and never shift the level. The first two literature shifts are, as their authors already noted, the same normal-ordering mechanism ($s_w = 1$ on fundamental fields). The third has been explained in the literature in three different ways: as normal ordering of the constraint, as the adjoint zero-point energy, and as a Jacobian. In the rule it is unambiguously an s_ε effect, namely the adjoint zero-point energy absorbed into the Calogero coupling, and not a Gauss-law effect. We test the rule on three phase spaces: $\text{Hilb}^n(\mathbb{C}^2)$ (where K is trivial), the Grassmannian (where K is non-trivial and the space is compact) and the Jordan quiver variety (where K is non-trivial and the space is non-compact). The tests use Molien integrals, Jeffrey–Kirwan residues, Atiyah–Bott sums and explicit time lattices for gauge group $U(n)$ with $n \leq 3$. We also show where the rule stops applying: once the Hamiltonian is quadratic in a generator, the continuum coherent-state path integral distorts the spectrum, and the time lattice converges only if the Fock space is truncated first.

1 Introduction

A first-order action $S = \oint \theta - \oint H dt$ with $d\theta = \omega$ defines a path-integral version of geometric quantisation. When H is the moment map of a torus action, the partition function $Z = \text{Tr } e^{-\beta H}$ on a circle is an equivariant index and can be written as a fixed-point sum. Gauged versions of this set-up appear in several places:

- the ADHM matrix model of the 4d quantum Hall effect [1], whose Higgs branch is the instanton moduli space ($\text{Hilb}^n(\mathbb{C}^2)$ in rank one);
- the Chern–Simons matrix model on the framed Jordan quiver [2, 4];
- the Polychronakos matrix model of the quantum Hall effect [3].

In each of these papers the relation between the bare Chern–Simons level and the line bundle that is actually quantised involves a shift.

*Claude Opus 5.5, an AI system developed by Anthropic. Almost all of this work, from the computations to the text, was generated by the AI; hence the first authorship (which alphabetical order happens to confirm).

What is already known. The shifts $k' \rightarrow k' - N$ [1] and $k + n \mapsto k$ [2] are both attributed by their authors to normal ordering of the fundamental-field bilinear in the trace of the Gauss law. The same “ordering issue” already appears in [4, 5], so the identification of these two shifts is not new. The shift $k \rightarrow k + 1$ of [3], which Polychronakos calls “a quantum reordering effect”, has been explained in three different ways:

- as the zero-point energy $N^2/2$ of the adjoint oscillators [6];
- as the comparison of ground-state energies, or equivalently a Vandermonde factor, between the matrix model and the Calogero model [7], and as a Jacobian (footnote in [1]);
- as “normal ordering a constraint” [8].

In addition, [1] lists it together with the Gauss-law shift. Two related questions are whether the expected half-integer ($K^{1/2}$ or spin^c) shifts appear [9–13], and whether the continuum coherent-state path integral can be trusted [14–18].

This paper organises these effects into one dictionary. Our main statements are:

1. *A rule* (Proposition 1). Each field carries two regularisation exponents, s_ε for its zero-point energy and s_w for its contribution to the Gauss law. The partition function is an overall factor times the normal-ordered index at a shifted level.
2. *A lemma* (Lemma 3). At each Jeffrey–Kirwan pole, the product of the weights of all matter modes equals the product of the tangent weights. Hence uniform symmetric regularisation computes $\chi(M, \mathcal{O}(k') \otimes K^{1/2})$.
3. *A no-go statement* (Proposition 2). Gauge-covariant time lattices have $s_w = 0$, so they never shift the level.
4. *A dictionary of the known shifts* (Table 1). The shifts of [1] and [2] are recovered as $s_w = 1$ on fundamental fields, in agreement with their authors. The shift of [3] is $s_\varepsilon = \frac{1}{2}$ on the adjoint: its Gauss law is normal ordered ($s_w = 0$), and the shift comes from the adjoint zero-point energy, whose off-diagonal part the Calogero coupling absorbs. This sides with [6, 7] and shows that attributing it to the ordering of the constraint [8], or grouping it with the Gauss-law shift [1], mixes up two different exponents.
5. *The limits of the rule* (Section 6). For Hamiltonians that are quadratic in a generator, the continuum path integral distorts the spectrum, and the time lattice has non-commuting limits.

Most individual entries are standard, and some identifications (items 4 and part of the Calogero discussion) are already in the literature, as summarised above. What we did not find in the literature, within the scope of our search, are:

- the two-exponent rule together with the criterion $s_\varepsilon = s_w = s$ for a K^s interpretation;
- the statement that gauge-covariant time lattices never shift the level;
- the JK-pole argument for the $K^{1/2}$ twist;
- the observation that on phase spaces with non-trivial K the midpoint lattice does not reproduce the half-form shift of the level;
- the non-commuting limits of the time lattice for non-linear Hamiltonians.

The Jeffrey–Kirwan (JK) residue machinery [19–21] is used throughout, mostly as a tool. Its application to the rank-one ADHM model (Section 7) adds only minor points to the general results of [19, 21]; see also [22].

2 Set-up

2.1 Models

Let $G = U(n)$ with gauge field A_t and bare Chern–Simons level k' , and let the matter consist of complex fields in representations of G with symplectic form $i d\bar{\phi} \wedge d\phi$. The action is

$$S = \int dt \left[\sum_{\text{fields}} i \bar{\phi} D_t \phi - k' \operatorname{tr} A_t - H \right], \quad (1)$$

where H is the moment map of a torus T (a linear combination of number operators). The Gauss law is $\mu_{\mathbb{R}} = k' \mathbf{1}$, so the FI parameter equals the bare level, $\zeta = k'$. A holomorphic constraint $\mu_{\mathbb{C}} = 0$, if present, is imposed by Koszul fermions χ with $Q = \sum (\mu_{\mathbb{C}})_{ab} \chi_{ba}$, $Q^2 = 0$. The partition function on S^1 with periodic boundary conditions reduces, after static diagonal gauge fixing, to a torus integral with Haar measure $\prod_{a \neq b} (1 - w_a/w_b)$. We consider three examples.

- **ADHM, rank one:** adjoints B_1, B_2 (weights t, q), fundamental I and antifundamental J , and $\mu_{\mathbb{C}} = [B_1, B_2] + IJ$. The Higgs branch is $M = \operatorname{Hilb}^n(\mathbb{C}^2)$. The scheme $\mu_{\mathbb{C}}^{-1}(0)$ is a reduced complete intersection [23], so imposing $\mu_{\mathbb{C}} = 0$ by Koszul fermions or by restricting to the variety gives the same character. We use the conventions of [24]: $T_{\mu} = t^{n(\mu)} q^{n(\mu')}$, $\mathcal{O}(1) = \wedge^n \mathcal{B}$, and the tangent weights $t^{1+l} q^{-a}$, $t^{-l} q^{1+a}$.
- **Grassmannian:** N fundamentals I_i with weights x_i and nothing else. Then $M = \operatorname{Gr}(n, N)$ and $K_M = \mathcal{O}(-N)$.
- **Framed Jordan quiver:** one adjoint Z (weight q) and f fundamentals (weights a_i). Then $M \cong \operatorname{Quot}^n(\mathbb{A}^{\oplus f}_1)$ [2].

2.2 Regularisation data

A single bosonic mode of weight $x = e^{-\beta\varepsilon}$ and holonomy w has the one-loop factor $1/\det(\partial_{\tau} + \varepsilon - ia)$. Different regularisations give $x^{s_{\varepsilon}} w^{s_w} / (1 - xw)$:

- normal ordering (equivalently, the forward Q-symbol time lattice) gives $s_{\varepsilon} = s_w = 0$;
- zeta-function regularisation with symmetric Matsubara cut-off gives $s_{\varepsilon} = s_w = \frac{1}{2}$;
- anti-normal ordering gives $s_{\varepsilon} = s_w = 1$;
- time lattices give $s_w = 0$ with $s_{\varepsilon} \in \{0, \frac{1}{2}, 1\}$ (Proposition 2).

Fermions contribute the inverse factor. In operator language, s_w is the ordering of the matter bilinear in the Gauss law, and s_{ε} is the ordering of H .

3 The rule

Proposition 1 (Rule). *Let $\delta = \sum_f q_f s_{w,f}$ be the sum, over charged fields, of the $U(1) \subset U(n)$ charge per colour times s_w . Let $X = \prod_{\text{modes}} x^{s_{\varepsilon}}$, with fermions contributing $x^{-s_{\varepsilon}}$. Then*

$$Z_{\text{reg}}(k') = X \cdot Z_{\text{normal}}(k' - \delta). \quad (2)$$

Proof. For adjoint fields the factors w^{s_w} cancel between (a, b) and (b, a) . For a field of charge q_f they give $\prod_a w_a^{q_f s_w}$, which combines with the Chern–Simons factor $\prod_a w_a^{-k'}$. What remains is the $x^{s_{\varepsilon}}$ factors, which are independent of w . $\square$

Proposition 2 (Gauge-covariant lattices). *Place the links U_j on the hopping terms, $\bar{z}_j(z_j - U_j z_{j-1})$, and evaluate H by any rule that is local in time and gauge covariant. Then the lattice determinant of a field in representation R is $\prod_{\omega} (a^M - \omega b^M)$, where ω runs over the eigenvalues of the holonomy in R . Hence $s_w = 0$ and the lattice never shifts the level. In the continuum limit, the forward Q -symbol, midpoint and same-time P -symbol prescriptions give $s_{\varepsilon} = 0, \frac{1}{2}, 1$ respectively.*

Proof. The kinetic matrix is block-cyclic with diagonal $a\mathbf{1}$ and sub-diagonal $-bR(U_j)$. Its determinant depends only on the conjugacy class of $R(U_M \cdots U_1)$. The values of a and b follow from expanding to first order in $\delta = \beta/M$. $\square$

Lemma 3 (Zeta function = $K^{1/2}$). *At every isolated JK pole, the product of the weights xw^{ρ} of all matter modes equals the product of the tangent weights of M at the corresponding fixed point. The reason is that the tangent character is the matter character minus the gauge character, and the gauge weights w_a/w_b multiply to one. Consequently, uniform zeta-function regularisation gives*

$$Z_{\zeta}(k') = \chi(M, \mathcal{O}(k') \otimes K_M^{1/2}). \quad (3)$$

More generally, Z_{reg} equals $\chi(M, \mathcal{O}(k') \otimes K_M^s)$ for all k' if and only if $s_{\varepsilon} = s_w = s$ for all fields.

Remark 4 (Global anomaly). If $\delta \notin \mathbb{Z}$, for example with an odd number of fundamentals and no antifundamentals in zeta-function regularisation, then Z_{ζ} is gauge invariant only for half-integer k' . This is the bosonic analogue of the fermionic global anomaly discussed in [19]. Lattices with integer level cannot reach the Dirac index in this case.

4 Three phase spaces

4.1 $\text{Hilb}^n(\mathbb{C}^2)$: trivial K

Because the field content contains both I and J , the level shift in zeta-function regularisation is $\delta = n(s_{w,I} - s_{w,J}) = 0$. Equivariantly $K \cong \mathcal{O} \otimes (qt)^n$, so the $K^{1/2}$ twist is only an overall factor $(qt)^{n/2}$, coming from the zero-point energy $E_0 = \frac{1}{2}(\sum_{\text{bos}} \varepsilon - \sum_{\text{ferm}} \varepsilon)$. The midpoint lattice ($s_w = 0$, $s_{\varepsilon} = \frac{1}{2}$) and the zeta function ($s_w = s_{\varepsilon} = \frac{1}{2}$) therefore agree on Hilb^n . The ordering in [1] (and the “ordering issue” of [5]) treats I anti-normally and J normally ($s_{w,I} = 1$, $s_{w,J} = 0$), which gives $k = k' - N$. This is not a power of K . If bosons and Koszul fermions, or B_1 and B_2 , are regularised differently, the overall factors are not powers of K either.

We checked the baseline $Z_{\text{normal}} = \chi(\text{Hilb}^n, \mathcal{O}(k))$ in four independent ways for $n \leq 4$, $0 \leq k \leq 3$:

- the Molien integral of [1];
- JK residues, where each colour-partition residue equals the corresponding Atiyah–Bott term (checked for $n \leq 5$);
- the Atiyah–Bott sum;
- the Hilbert series of products of alternants [25].

For gauge-covariant lattices with $n = 2, 3$, we computed every determinant numerically (random $U(n)$ links for the invariance check, and holonomy integrals for Z). The results converge as $O(1/M)$ to $X \cdot Z_{\text{normal}}(k')$ with no level shift.

4.2 $\text{Gr}(n, N)$: non-trivial K , compact

Here $\delta = N s_w$, and the rule gives:

- normal ordering: $s_{(k'^n)}(x) = \chi(\mathcal{O}(k'))$;
- zeta function: $(\det x)^{n/2} s_{((k' - N/2)^n)}(x) = \chi(\mathcal{O}(k') \otimes K^{1/2})$, a level shift by $N/2$;
- anti-normal ordering: $\chi(\mathcal{O}(k') \otimes K)$, a level shift by N ;
- every gauge-covariant lattice, including the midpoint rule: $(\det x)^{sn} s_{(k'^n)}$.

The naive identification “midpoint rule = Weyl ordering = metaplectic correction” is therefore correct for the zero-point part and wrong for the level part. On Hilb^n the failure is invisible because K is trivial. This does not contradict the semiclassical statements that the Weyl symbol removes the Solari–Kochetov correction [9], or that the metaplectic correction is tied to the correct continuum limit of coherent-state path integrals [10]. Those statements concern the continuum or the semiclassical propagator. Ours concerns the Chern–Simons level in a gauge-covariant time discretisation, where the half-form shift must be put in by hand through the bare level. For half-form and metaplectic-c quantisation with torus actions see [11, 12]; in the quantum-Hall context the metaplectic correction is sometimes deliberately omitted [13]. We checked this for $(n, N) \in \{(1, 2), (1, 3), (2, 3), (2, 4)\}$ and $k \leq 3$.

4.3 Jordan quiver: non-trivial K , non-compact

In normal ordering the Molien integral and the JK residues reproduce Theorem A of [2] (which notes that the formula first appeared in [4]),

$$\text{ch } \mathcal{H} = \tilde{H}_{(k^n)}(a; q) \prod_{i=1}^n \frac{1}{1 - q^i}, \quad (4)$$

where $\tilde{H}_\mu$ is built from Jing operators as in that reference. We checked this for $f \leq 2$, $n \leq 3$ and $k \leq 3$. In zeta-function regularisation the JK residues multiplied by (product of tangent weights) $^{1/2}$ reproduce Z_ζ , which confirms Lemma 3 here with $K \cong \mathcal{L}_{\det}^{-f}$ and a level shift of $f/2$. The shift $k + n \mapsto k$ of [2] is $s_w = 1$ on fundamentals. For $f = 1$, the Calogero model obtained from the Hermitian (real-polarised) version [3] with coupling $\ell(\ell - 1)$, $\ell = k + 1$, satisfies

$$Z_{\text{Cal}}(\ell = k + 1) = q^{n^2/2} a^{-kn} Z_{\text{normal}}(k), \quad (5)$$

which we checked for $n = 2$ by finite differences. The factor $q^{n^2/2}$ is the zero-point energy of the adjoint ($s_\varepsilon = \frac{1}{2}$). Its off-diagonal part $q^{n(n-1)/2}$ is the degree of the Vandermonde factor, which the Calogero coupling absorbs. Thus the shift of [3] is an s_ε effect, not an s_w effect. The relation to the zero-point energy is not new: [6] identify the $N^2/2$ zero-point term as the cause of the shift, and [7] compare the ground-state energies $N^2/2$ and $N/2$ of the matrix and Calogero models. What the rule adds is the following. Polychronakos’s Gauss law is normal ordered, so it has $s_w = 0$. Calling the effect a “level shift” is therefore a matter of which variables one uses (the Calogero coupling), not a Gauss-law effect. In particular, it should not be identified with the shifts of [1, 2], contrary to the grouping in [1] and the attribution in [8].

5 The dictionary

6 Limits of the rule

The rule assumes that H is linear in number operators. For $H = \varepsilon N + \lambda N^2$ (the one-mode sector of $\text{tr}(B^\dagger B)^2$), a Hubbard–Stratonovich transformation evaluates the continuum coherent-state path integral exactly. For a quadratic symbol h it gives $\sum_{m \geq 0} e^{-\beta h(m+1/2)}$.

Reported shift	Model	Explanation in the literature	Position in the rule	Geometric meaning
$k' \rightarrow k' - N$ [1]	ADHM	normal ordering of $\varphi\varphi^\dagger$ [1, 5]	$s_{w,I} = 1, s_{w,J} = 0$	not a power of K
$k + n \mapsto k$ [2]	Jordan quiver	normal ordering of (BA) [2]	$s_w = 1$ on fundamentals	K^1 (non-equivariantly)
$k \rightarrow k + 1$ [3]	Jordan quiver, $f = 1$	reordering [3]; zero point [6, 7]; Jacobian [1]; constraint ordering [8]	adjoint $s_\varepsilon = \frac{1}{2}, s_w = 0$	overall factor, read as Calogero coupling
$\text{spin}^c / K^{1/2}$	all	half-form correction	$s_\varepsilon = s_w = \frac{1}{2}$ for all fields	level shift iff K non-trivial
gauge-covariant lattice	all	—	$s_w = 0$	no level shift

Table 1: The dictionary of quantum shifts.

- With the Q-symbol, the energies become $E_m + \varepsilon/2 + 3\lambda/4 + 2\lambda m$. The shift depends on m , so the spectrum is distorted.
- With the Weyl symbol the shift is the constant $-\lambda/4$. This is an accident of N^2 .

The Q-symbol time lattice equals $\text{Tr}(:e^{-\delta h}:)^M$. It converges as $O(1/M)$ on a truncated Fock space. At any finite M , however, the transfer-matrix eigenvalues exceed one for $m \gtrsim 3.7M$, so the full trace diverges. This refines the statement of [14] that the time-discretised path integral does not suffer from the breakdown: the order of the limits matters. For proposed consistent continuum formulations and the ensuing debate, see [15–18]. We do not take a position on that debate. Our statement concerns only the lattice.

7 Remarks on JK residues and wall crossing

For rank-one ADHM we also verified the following.

- The fixed-contour Molien integral selects the chamber $\text{sign}(\zeta) = \text{sign}(k)$, consistently with $\zeta = k'$.
- The wall-crossing difference is

$$Z_+ - Z_- = \chi(\mathcal{O}(k)) - (qt)^{-(k+1)n} \chi(\mathcal{O}(k))^\vee. \quad (6)$$

It vanishes at $k = 0$ and factorises at $k = 1$ because $\chi(\mathcal{O}(1)) = \chi(\mathcal{O}(-1))$, which we observed for $n \leq 5$.

- In rank two there is no wall crossing for $|k| < N$.

These follow from the general results of [19, 21]. A case analysis of the background charge already appears in [22].

8 Conclusions

One rule with two exponents per field organises the quantum shifts of first-order gauged matrix models. Part of the resulting dictionary confirms identifications already made by the original authors [1, 2, 6, 7]. Its use is to make the remaining distinctions sharp and to resolve conflicting attributions. It separates three kinds of shift: Gauss-law (dictionary) shifts (s_w), zero-point and

Calogero-type shifts (s_ε), and genuine spin^c twists ($s_\varepsilon = s_w = \frac{1}{2}$). The comparison between trivial and non-trivial canonical bundles is essential: on Hilb^n several inequivalent prescriptions cannot be told apart. A natural next step is a phase space with both non-trivial K and a holomorphic constraint, together with non-linear Hamiltonians in the gauged setting.

A Scope of the numerical checks

All checks are automated tests (346 in total). Power series in q, t are compared up to total degree 10–16, and numerical integrals are compared to relative accuracy 10^{-4} – 10^{-12} . The ranges are:

- rank-one ADHM: $n \leq 4$, $0 \leq k \leq 3$; JK residues for $n \leq 5$;
- lattices: $n \leq 3$;
- rank two: $n \leq 2$;
- Grassmannians: $n \leq 2$, $N \leq 4$;
- Jordan quiver: $n \leq 3$, $f \leq 2$.

For $k \geq 2$, the Hilbert series of products of alternants is computed probabilistically, from ranks modulo a large prime.

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